Circular and Hyperbolic: Trigonometric Functions, Their Hyperbolic Twins, and Their Inverses

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1 One function, two directions

The circular functions (sin, cos, tan) and the hyperbolic functions (sinh, cosh, tanh) are closely related, but neither real family is a special case of the other as a real-valued function. Extend their exponential formulas to complex arguments and a quarter-turn of the argument converts one family into the other. Circular sine oscillates on the real axis; on the imaginary axis it grows like hyperbolic sine.

Many connections follow from this fact: conversion identities, logarithmic formulas for the inverses, and paired integrals of calculus. There are also real-variable connections through differential equations and area. One can understand those first; complex analysis explains why the formulas continue to hold after an imaginary substitution and how to handle their branches.

The essay starts with the exponential definitions, derives the conversion identities, examines their geometry, inverts both families, and ends with the consequences for calculus.

2 The exponential definitions

Euler's formula ties the circular functions to the exponential:

$$e^{iz} = \cos z + i \sin z.$$

Write it at z and at $-z$, then add or subtract the two equations. This gives sine and cosine as combinations of e^{iz} and e^{-iz}:

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

The hyperbolic functions are the same combinations with the i removed:

$$\cosh z = \frac{e^{z} + e^{-z}}{2}, \qquad \sinh z = \frac{e^{z} - e^{-z}}{2}.$$

In both families cos and cosh are the even parts of an exponential, and sin and sinh are the odd parts, up to a factor of i. Tangent, cotangent, secant and the rest are quotients of these, so they inherit the same structure.

These formulas define all four functions on all of $\mathbb{C}$ as entire functions. Their power series match that picture. $\sinh z = z + z^3/3! + z^5/5! + \cdots$ has the same coefficients as $\sin z$, without the alternating signs. The sign pattern in the sine series is exactly what substituting iz produces.

3 A real-variable bridge: differential equations

The two families can be compared before introducing complex numbers. On the real line, their defining pairs satisfy

$$\frac{d}{dt}\begin{pmatrix}\cos t\\ \sin t\end{pmatrix} = \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\begin{pmatrix}\cos t\\ \sin t\end{pmatrix}, \qquad \frac{d}{dt}\begin{pmatrix}\cosh t\\ \sinh t\end{pmatrix} = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}\cosh t\\ \sinh t\end{pmatrix}.$$

Both start at $(1, 0)$. The sign in the upper-right corner distinguishes a rotation from a hyperbolic rotation. In particular, each coordinate solves a second-order equation:

$$y'' = -y \quad \text{for } \sin, \cos, \qquad y'' = y \quad \text{for } \sinh, \cosh .$$

This is the oscillation-versus-growth distinction in its simplest form. The corresponding addition laws are matrix multiplication laws:

$$\begin{pmatrix}\cos(s+t) & -\sin(s+t)\\ \sin(s+t) & \cos(s+t)\end{pmatrix} = R(s)R(t), \qquad \begin{pmatrix}\cosh(s+t) & \sinh(s+t)\\ \sinh(s+t) & \cosh(s+t)\end{pmatrix} = H(s)H(t),$$

where R and H denote the displayed matrices at their respective arguments. Both have determinant 1; direct multiplication shows that they preserve $x^2 + y^2$ and $x^2 - y^2$, respectively. This gives the circle and hyperbola identities without complex analysis.

The imaginary substitution explains the changed sign: if $f'' = -f$ and $g(t) = f(it)$, the chain rule gives $g''(t) = -f''(it) = g(t)$. This short argument uses the complex extension of f, but it reveals why $i^2 = -1$ interchanges the two differential equations.

4 The rotation identities

Substitute iz into the circular definitions. Then $e^{i\cdot iz} = e^{-z}$, and the circular formulas become the hyperbolic ones:

$$\cos(iz) = \cosh z, \qquad \sin(iz) = i\sinh z, \qquad \tan(iz) = i\tanh z.$$

In the other direction:

$$\cosh(iz) = \cos z, \qquad \sinh(iz) = i\sin z, \qquad \tanh(iz) = i\tan z.$$

The form $\sinh x = -i\sin(ix)$ is the first sine identity solved for sinh. The reciprocal functions follow by division: $\sec(iz) = \operatorname{sech} z$, $\csc(iz) = -i\operatorname{csch} z$, $\cot(iz) = -i\coth z$.

The even functions (cos, cosh, sec, sech) carry over unchanged. The odd ones pick up a factor of i. That is forced: multiplying the argument by i multiplies the n-th power-series term by i^n. For even n this gives ± 1, which cancels against the alternating signs. For odd n it leaves one extra i.

Combined with the addition formulas, these identities split any circular function into real and imaginary parts:

$$\begin{aligned}\sin(x+iy) &= \sin x\cosh y + i\cos x\sinh y,\\ \cos(x+iy) &= \cos x\cosh y - i\sin x\sinh y.\end{aligned}$$

One consequence is $|\sin(x+iy)|^2 = \sin^2 x + \sinh^2 y$. So sine is bounded on the real line but grows like $e^{|y|}/2$ off it. Liouville's theorem requires this, since a bounded entire function must be constant.

A reliable translation rule. For identities in sine and cosine, substitute iz for each argument, replace $\sin(iz)$ by $i\sinh z$ and $\cos(iz)$ by $\cosh z$, and simplify the powers of i. Quotients can be handled the same way after writing out their numerators and denominators. Some examples:

Circular	Hyperbolic
$\cos^2 x + \sin^2 x = 1$	$\cosh^2 x - \sinh^2 x = 1$
$\cos 2x = 1 - 2\sin^2 x$	$\cosh 2x = 1 + 2\sinh^2 x$
$1 + \tan^2 x = \sec^2 x$	$1 - \tanh^2 x = \operatorname{sech}^2 x$
$\sin(a+b) = \sin a \cos b + \cos a \sin b$	$\sinh(a+b) = \sinh a \cosh b + \cosh a \sinh b$

This is an identity substitution, not a mnemonic about signs: two sine factors contribute $i^2 = -1$, while four contribute $i^4 = 1$.

5 Geometry: the circle and the hyperbola

The point $(\cos t, \sin t)$ lies on the unit circle $x^2 + y^2 = 1$. The point $(\cosh t, \sinh t)$ lies on the right branch of the unit hyperbola $x^2 - y^2 = 1$. Substituting $y \mapsto iy$ turns one curve into the other, which is the rotation identity seen in coordinates.

The common meaning of t is signed area, not arc length. Sweep the ray from the origin along the curve from parameter 0 to t, counting orientation and repeated turns. The swept area is $t/2$ in both cases. For a curve parameterized by $(x(s), y(s))$, the area swept from the origin is

$$A = \frac{1}{2}\int_0^t \big(x(s)\,y'(s) - y(s)\,x'(s)\big)\,ds.$$

For the circle the integrand is $\cos^2 + \sin^2 = 1$. For the hyperbola it is $\cosh^2 - \sinh^2 = 1$. Both give $A = t/2$.

On the unit circle, the arc length is $|t|$, twice the unsigned sector area. On the hyperbola, arc length is not $|t|$. That is why "hyperbolic angle" is defined by twice the signed swept area. It is also why the hyperbolic angle is unbounded, while circular angle wraps around after 2π.

This shows the main structural difference between the two families. The circle is compact, so the circular functions are periodic on $\mathbb{R}$. The hyperbola is not, so the hyperbolic functions are injective or monotone on suitable pieces of $\mathbb{R}$. In $\mathbb{C}$ the periods reappear rotated: sinh and cosh have period $2\pi i$, and tanh has period πi.

6 Why the inverses are logarithms

Each of the six functions is a rational function of a single exponential: $u = e^w$ for the hyperbolic family, $u = e^{iw}$ for the circular one. To solve $f(w) = z$, rewrite it as a polynomial equation in u, solve for u, and take a logarithm. For sin, cos, sinh and cosh the polynomial is quadratic. For tan and tanh it is linear in u^2.

arcsinh. $\sinh w = z$ becomes $u - 1/u = 2z$, that is $u^2 - 2zu - 1 = 0$. The roots multiply to -1, so for real z exactly one is positive, namely $z + \sqrt{z^2+1}$:

$$\operatorname{arcsinh} z = \ln\left(z + \sqrt{z^2+1}\right).$$

arccosh. $\cosh w = z$ becomes $u^2 - 2zu + 1 = 0$. The roots multiply to $+1$, so they are reciprocals and give $\pm w$. That is the evenness of cosh. Taking the larger root, for real $z \geq 1$:

$$\operatorname{arccosh} z = \ln\left(z + \sqrt{z^2 - 1}\right).$$

arctanh. $\tanh w = (u^2 - 1)/(u^2 + 1) = z$, so $u^2 = (1 + z)/(1 - z)$:

$$\operatorname{arctanh} z = \frac{1}{2} \ln \frac{1 + z}{1 - z}.$$

The circular inverses. These come out the same way with $u = e^{iw}$, and $w = -i \ln u$ at the end:

$$\begin{aligned}
\arcsin z &= -i \ln\left(iz + \sqrt{1 - z^2}\right), \\
\arccos z &= -i \ln\left(z + i\sqrt{1 - z^2}\right), \\
\arctan z &= \frac{1}{2i} \ln \frac{1 + iz}{1 - iz}.
\end{aligned}$$

For real $z \in [-1, 1]$, the two roots of the arcsin quadratic are $iz \pm \sqrt{1 - z^2}$, and both lie on the unit circle. Choosing the $+$ root puts w in $[-\pi/2, \pi/2]$. The other root gives $\pi - w$, the second real solution of $\sin w = z$. The periodicity 2π comes from the log, which is defined only up to $2\pi i$; multiplied by $-i$, that becomes 2π in w.

The pattern is general. Every inverse trigonometric and inverse hyperbolic function is a logarithm of an algebraic function of z. Nothing new is needed beyond log and the square root.

7 Converting between the inverses

Inverting the rotation identities gives conversion rules for the inverse functions. Start from $\sin(iw) = i \sinh w$ and let $w = \operatorname{arcsinh} z$. Then $\sin(iw) = iz$, so $iw = \arcsin(iz)$. The same argument run from the other identity gives the second formula, and tan works the same way:

Inverse rule	Comes from
$\arcsin z = -i \operatorname{arcsinh}(iz)$	$\sinh(iw) = i \sin w$
$\operatorname{arcsinh} z = -i \arcsin(iz)$	$\sin(iw) = i \sinh w$
$\arctan z = -i \operatorname{arctanh}(iz)$	$\tanh(iw) = i \tan w$
$\operatorname{arctanh} z = -i \arctan(iz)$	$\tan(iw) = i \tanh w$
$\operatorname{arccosh} z = \pm i \arccos z$	$\cosh(iw) = \cos w$

These hold for the principal branches. The cosine pair is the exception. Because cos and cosh are even, inverting only fixes iw up to sign. The principal arccosh has non-negative real part, so the sign is $+i$ for $\Im z > 0$ and $-i$ for $\Im z < 0$. On the real segment it depends on how the cut is closed. For example, $\operatorname{arccosh}(1/2) = i\pi/3 = i \arccos(1/2)$.

The two routes to the logarithmic form now agree. Substituting $u = iz$ into $\operatorname{arcsinh} u = \ln\left(u + \sqrt{u^2 + 1}\right)$ gives

$$\arcsin z = -i \operatorname{arcsinh}(iz) = -i \ln\left(iz + \sqrt{1 - z^2}\right).$$

This is the same formula the direct derivation produced. So there is one logarithmic formula per pair of functions, not per function. The rotation identity transports it from one member of the pair to the other.

8 Branches, cuts, and where the identities fail

Every formula above contains two multivalued operations, a square root and a logarithm. Choosing principal branches makes the inverses single-valued, at the cost of a cut. The standard principal cuts are:

Function	Cut	Values (strip)
arcsin, arccos	$(-\infty, -1] \cup [1, \infty)$	$\Re w \in [-\pi/2, \pi/2]$ and $[0, \pi]$
arctan	$(-i\infty, -i] \cup [i, i\infty)$	$\Re w \in (-\pi/2, \pi/2)$
arcsinh	$(-i\infty, -i] \cup [i, i\infty)$	$\Im w \in [-\pi/2, \pi/2]$
arctanh	$(-\infty, -1] \cup [1, \infty)$	$\Im w \in (-\pi/2, \pi/2)$
arccosh	$(-\infty, 1]$	$\Re w \geq 0$

The rotation $z \mapsto iz$ turns the real cuts of arcsin into the imaginary cuts of arcsinh. This is the inverse identity of the previous section, seen in the cut geometry.

The identities fail in three typical ways.

1. **Inverse after function is not the identity.** $\arcsin(\sin z) = z$ only in the principal strip. For example, $\arcsin(\sin \pi) = 0$.
2. **Log rules break.** $\ln(ab) = \ln a + \ln b$ holds only up to $2\pi i$. So the arctanh formula cannot be split as $\frac{1}{2}\big(\ln(1+z) - \ln(1-z)\big)$ without checking the branches.
3. **The square root moves the cut.** $\sqrt{z^2-1}$ and $\sqrt{z-1}\sqrt{z+1}$ agree when $\Re z > 0$ and differ by sign when $\Re z < 0$. The naive arccosh formula therefore gives the wrong branch near the negative axis. At $z = -2$, the naive argument $-2 + \sqrt{3}$ has modulus $2 - \sqrt{3} < 1$. Any logarithm of it has negative real part $\ln(2-\sqrt{3})$, whereas a principal arccosh boundary value has positive real part $\ln(2+\sqrt{3})$. A formula consistent on the slit plane needs the factored root:
$$\operatorname{arccosh} z = \ln\big(z + \sqrt{z+1}\,\sqrt{z-1}\big)\,.$$

The working rule: begin on a region where the branch choices are unambiguous, then extend an identity only while both sides remain analytic on the same connected region. Crossing a cut requires new branch information. Kahan's paper on branch cuts, *Much Ado About Nothing's Sign Bit*, is the standard reference for getting these right in floating point.

9 Calculus consequences

The derivatives of the inverses fall into pairs that differ only in a sign under the root or in the denominator:

Function	Derivative	Real domain
$\arcsin x$	$1/\sqrt{1-x^2}$	$\lvert x\rvert < 1$
$\operatorname{arcsinh} x$	$1/\sqrt{1+x^2}$	$x \in \mathbb{R}$
$\operatorname{arccosh} x$	$1/\sqrt{x^2-1}$	$x > 1$
$\arctan x$	$1/(1+x^2)$	$x \in \mathbb{R}$
$\operatorname{arctanh} x$	$1/(1-x^2)$	$\lvert x\rvert < 1$

Read backwards, this explains why integrals that look alike produce such different-looking answers. $\int dx/(1-x^2)$ can be done by partial fractions, which yields $\frac{1}{2}\ln\lvert(1+x)/(1-x)\rvert$ on any interval

avoiding ± 1. On $(-1, 1)$ this is $\operatorname{arctanh} x$. $\int dx/(1+x^2)$ has no real partial fractions, so the answer gets a new name, arctan. Over $\mathbb{C}$ it factors as $(1+ix)(1-ix)$, and partial fractions give exactly the logarithmic formula for arctan from earlier. The "new" function is an artifact of working over $\mathbb{R}$.

The same holds for the substitution families of integration:

Integrand contains	Circular substitution	Hyperbolic substitution
$\sqrt{a^2 - x^2}$	$x = a \sin t$	$x = a \tanh t$
$\sqrt{a^2 + x^2}$	$x = a \tan t$	$x = a \sinh t$
$\sqrt{x^2 - a^2}$	$x = a \sec t$	$x = a \cosh t$

The hyperbolic column often gives the answer directly in log form. The circular column produces things like $\int \sec t \, dt = \ln|\sec t + \tan t|$, which is a hyperbolic inverse in disguise.

The Gudermannian. A real-variable bridge between the families, with no i involved, is the Gudermannian function:

$$\operatorname{gd} x = \int_0^x \operatorname{sech} t \, dt = \arctan(\sinh x) = 2 \arctan\Big(\tanh \frac{x}{2}\Big).$$

It satisfies $\sinh x = \tan(\operatorname{gd} x)$, $\cosh x = \sec(\operatorname{gd} x)$ and $\tanh x = \sin(\operatorname{gd} x)$. Thus it links coordinates on the hyperbola to trigonometric ratios at a real circular angle. For $|\theta| < \pi/2$ its inverse is

$$\operatorname{gd}^{-1} \theta = \int_0^\theta \sec u \, du = \operatorname{arcsinh}(\tan \theta).$$

The integral also gives the vertical coordinate of the spherical Mercator projection as a function of latitude θ.